On the Width Inference Problem of FIRRTL Programs

Abstract. In this note, we investigate the width inference problem of FIRRTL programs. We first formulate the width inference problem of FIRRTL programs as the problem of finding the minimum solution of linear inequality systems of a special form (denoted by FIRLIS). Moreover, we show that for each satisfiable linear inequality system from FIRLIS, the minimum solution always exists. Finally, we present a procedure for computing the minimum solution of a given FIRLIS constraint.

1 FIRLIS constraints

Let $x_1, \cdots, x_n$ be variables that range over $\mathbb{N}$, i.e. the set of natural numbers. The width inference problem of FIRRTL programs can be formulated as finding the minimum solution of a formula φ that is a conjunction of linear inequalities of the form

$$x_i \geqslant a_0 + \sum_{j \in [n]} a_j x_j,$$

where $a_0 \in \mathbb{Z}$ and $a_j \in \mathbb{N}$. Let us use FIRLIS to denote the class of formulas φ of this form.

Definition 1 (FIRLIS constraints). *A linear constraint φ is called a FIRLIS constraint if it is of the form $\varphi_1 \wedge \varphi_2$, where φ_1 is a conjunction of inequalities of the form*

$$x_i \geqslant a_0 + \sum_{j \in [n]} a_j x_j,$$

such that $a_0 \in \mathbb{Z}$ and $a_j \in \mathbb{N}$, and φ_2 is a conjunction of inequalities of the form $b_0 \geqslant \sum_{i \in [n]} b_i x_i$ where $b_i \in \mathbb{N}$ for each $i \in [0, n]$. The equalities $x_i \geqslant a_0 + \sum_{j \in [n]} a_j x_j$ are called the inequalities of x_i.

Remark 1. Note here for technical convenience, we additionally include the inequalities of the form $b_0 \geqslant \sum_{i \in [n]} b_i x_i$ in FIRLIS constraints.

For $n \geqslant 1$, let $\preceq_n$ denote the partial order over $\mathbb{N}^n$ such that $(u_1, \cdots, u_n) \preceq_n (v_1, \cdots, v_n)$ iff for each $i \in [n]$, $u_i \leqslant v_i$.

For a FIRLIS constraint φ, a solution $(u_1, \cdots, u_n)$ is said to be the *minimum solution* of φ if for each solution $(v_1, \cdots, v_n)$ of φ, we have $(u_1, \cdots, u_n) \preceq_n (v_1, \cdots, v_n)$.

Example 1. Consider the FIRLIS constraint $\varphi \equiv x_1 \geqslant 2x_2 - 3 \wedge x_2 \geqslant 2x_1 + 1$. Then the minimum solution of φ is $(0, 1)$. On the other hand, if $\varphi \equiv x_1 \geqslant 2x_1 \wedge x_1 \geqslant 1$, then φ is unsatisfiable, thus the minimum solution of φ does not exist.

2 Existence of minimum solutions of FIRLIS

Proposition 1. *The minimum (nonnegative) solution of a satisfiable FIRLIS constraint alway exists.*

Proof. Let $\varphi = \varphi_1 \wedge \varphi_2$ be a FIRLIS constraint such that φ is a conjunction of inequalities of the form

$$\bigwedge_{i\in[n]} x_i \geqslant a_0 + \sum_{j\in[n]} a_j x_j,$$

where $a_0 \in \mathbb{Z}$ and $a_j \in \mathbb{N}$, and φ_2 is a conjunction of the inequalities of the form $b_0 \geqslant \sum_{i\in[n]} b_i x_i$.

We show the following claim.

Claim. *Let $(u_1, \cdots, u_n)$ and $(v_1, \cdots, v_n)$ be two solutions of φ and*

$$(w_1, \cdots, w_n) = (\min(u_1, v_1), \cdots, \min(u_n, v_n)).$$

Then $(w_1, \cdots, w_n)$ is also a solution of φ.

Proof of the claim. Let I_1 denote the set of $i \in [n]$ such that $u_i \leqslant v_i$. Then for each $i \in I_1$, $w_i = u_i$, and for each $i \in [n]\backslash I_1$, $w_i = v_i$.

For each $i \in [n]$, we show that $w_i \geqslant a_{i,0} + \sum_{j\in[n]} a_{i,j} w_j$.

Evidently, for each inequality of the form $b_0 \geqslant \sum_{i\in[n]} b_i x_i$ in φ_2, we have $b_0 \geqslant \sum_{i\in[n]} b_i u_i \geqslant \sum_{i\in[n]} b_i w_i$.

Let us consider an inequality of φ_1, say $x_i \geqslant a_0 + \sum_{j\in[n]} a_j x_j$ with $a_0 \in \mathbb{Z}$ and $a_j \in \mathbb{N}$.

– If $u_i \leqslant v_i$, then

$$\begin{aligned} w_i = u_i &\geqslant a_0 + \sum_{j\in[n]} a_j u_j \\ &= a_0 + \sum_{j\in I_1} a_j u_j + \sum_{j\in[n]\backslash I_1} a_j u_j \\ &\geqslant a_0 + \sum_{j\in I_1} a_j u_j + \sum_{j\in[n]\backslash I_1} a_j v_j \\ &= a_0 + \sum_{j\in I_1} a_j w_j + \sum_{j\in[n]\backslash I_1} a_j w_j \\ &= a_0 + \sum_{j\in[n]} a_j w_j. \end{aligned}$$

– If $u_i > v_i$, then

$$\begin{aligned} w_i = v_i &\geqslant a_0 + \sum_{j\in[n]} a_j v_j \\ &= a_0 + \sum_{j\in I_1} a_j v_j + \sum_{j\in[n]\backslash I_1} a_j v_j \\ &\geqslant a_0 + \sum_{j\in I_1} a_j u_j + \sum_{j\in[n]\backslash I_1} a_j v_j \\ &= a_0 + \sum_{j\in I_1} a_j w_j + \sum_{j\in[n]\backslash I_1} a_j w_j \\ &= a_0 + \sum_{j\in[n]} a_j w_j. \end{aligned}$$

Therefore, $(w_1, \cdots, w_n)$ is a solution of φ.

The proof of the claim is complete.

From the claim, we deduce that if φ is satisfiable, then the minimum (nonnegative) solution exists. □

3 Computation of the minimum solution

In this section, we present a procedure to compute the minimum solution of a FIRLIS constraint. The procedure utilizes the concept of dependency graphs.

Definition 2 (Dependency graph). *Let φ be a FIRLIS constraint over $x_1, \cdots, x_n$ such that there is $B \in \mathbb{N}$ such that for every $i \in [n]$, the number of inequalities of x_i is at most B. For each $i \in [n]$, let us assign a label $l \in [B]$ to each inequality of x_i. For convenience, we write an equality $x_i \geqslant a_0 + \sum_{j \in [n]} a_j x_j$ of label l as $l : x_i \geqslant a_0 + \sum_{j \in [n]} a_j x_j$. The dependency graph of φ, denoted by G_φ, is defined as (V_φ, E_φ), where $V_\varphi = \{x_1, \cdots, x_n\}$ and $E_\varphi \subseteq V_\varphi \times [B] \times \mathbb{N} \times V_\varphi$ comprises the edges (x_i, l, a_j, x_j) such that φ contains an inequality $x_i \geqslant a_0 + \sum_{j \in [n]} a_j x_j$ of label l with $a_j > 0$.*

Note that G_φ is an edge-labeled and edge-weighted directed graph.

In the sequel, we shall present a procedure to decide the satisfiability of FIRLIS constraints and compute the minimum solutions for satisfiable formulas.

Let us first consider the special cases that the dependency graphs are acyclic or strongly connected. Later on, we shall consider the more general case.

G_φ is acyclic. We compute the minimum solution $(u_1, \cdots, u_n)$ as follows.

1. Compute a topological sorting of G_φ, say $x_{i_1}, \cdots, x_{i_n}$.
2. Each inequality for x_{i_n} is of the form $x_{i_n} \geqslant a_0$, since it does not depends on the other variables. Let $a'_{i_n,0}$ denote the maximum of these a_0. (Without loss of generality, we assume that there is at least such a_0, since we can always add the inequality $x_{i_n} \geqslant 0$.)
3. Let $j := n$. We iterate the following procedure until $j = 0$.
   (a) Let u_{i_j} be $\max(a'_{i_j,0}, 0)$.
   (b) If $j > 1$, then do the following: Replace each occurrence of x_{i_j} in the right-hand sides of the inequalities for $x_{i_1}, \cdots, x_{i_{j-1}}$ by u_{i_j} and simplify the resulting expressions (by evaluating the subexpressions involving only the constants). Then each inequality for $x_{i_{j-1}}$ is of the form $x_{i_{j-1}} \geqslant a_0$. Let $a'_{i_{j-1},0}$ be the maximum of these a_0.
   (c) Let $j := j - 1$.

G_φ is strongly connected. We first prove the following result.

Proposition 2. *Suppose that G_φ is strongly connected. If G_φ contains an edge of weight greater than 1 or two distinct edges out of some vertex x_i of the same label, then there is a constant $C \in \mathbb{N}$ such that the values of the variables in every solution of φ are bounded by C. As a result, the number of nonnegative solutions of φ is finite.*

Proof. Suppose that G_φ is strongly connected.

Let us first consider the case that G_φ contains an edge of weight greater than 1, say (x_i, l, a, x_j) with $a > 1$. Take an arbitrary vertex x_k. Then there is a path π_1 from x_k to x_i, and a path π_2 from x_j to x_k, since G_φ is strongly connected. Therefore, the path π_0 obtained by concatenating π_1, (x_i, l, a, x_j), and π_2 is a cycle from x_k to x_k that traverses the edge (x_i, l, a, x_j). Let $\pi_0 = (z_0, l_1, a_1, z_1)(z_1, l_2, a_2, z_2) \cdots (z_{m-1}, l_m, a_m, z_m)$, where $z_0 = z_m = x_k$. Moreover, for each $r \in [m]$, let the inequality of z_{r-1} labeled by l_r be $z_{r-1} \geqslant b_{r-1,0} + \sum_{j' \in [n]} b_{r-1,j'} x_{j'}$. Then we compute the equalities $Ineq_r$ for $r \in [m]$ inductively as follows: Let $Ineq_1$ be $z_0 \geqslant b_{0,0} + \sum_{j' \in [n]} b_{0,j'} x_{j'}$. For each $r \in [m-1]$, $Ineq_{r+1}$ is obtained from $Ineq_r$ by replacing z_r with $b_{r,0} + \sum_{j' \in [n]} b_{r,j'} x_{j'}$ and

simplifying the resulting inequality. Then the inequality $Ineq_m$ is of the form $z_0 \geqslant c_0 + \sum_{j' \in [n]} c_{j'} x_{j'}$. From the inductive computation of $Ineq_m$ and the fact that $z_0 = z_m = x_k$, we know that $c_k = \prod_{j' \in [m]} a_{j'}$. Because (x_i, l, a, x_j) is a path on π_0 and $a > 1$, we deduce that $c_k > 1$. Therefore, $-c_0 \geqslant (c_k - 1)x_k + \sum_{j' \in [n], j' \neq k} c_{j'} x_{j'}$. This means that if φ is satisfiable, then the value of x_k is bounded by $-c_0$.

Consequently, in the case that G_φ contains an edge of weight greater than 1, there is a constant C such that the values of variables in every solution are bounded by C.

Then let us consider the case that G_φ contains two distinct edges out of some vertex x_i of the same label, say (x_i, l, a_1, x_{j_1}) and (x_i, l, a_2, x_{j_2}) (where $a_1, a_2 \geqslant 1$ and $j_1 \neq j_2$). Let the inequality of x_i labeled by l be $x_i \geqslant a'_0 + \sum_{j' \in [n]} a'_{j'} x_{j'}$. Then $a'_{j_1} = a_1$ and $a'_{j_2} = a_2$. Take an arbitrary vertex x_k. Then there is a path π_0 from x_k to x_i. Moreover, there are paths π_1 and π_2 from x_{j_1} to x_k and from x_{j_2} to x_k respectively. Let π'_1 be the concatenation of π_0, (x_i, l, a_1, x_{j_1}). Then π'_1 is a path from x_k to x_k. Similarly, let π'_2 be the concatenation of π_0, (x_i, l, a_2, x_{j_2}). Then π'_2 is another path from x_k to x_k. Let the inequality corresponding to the first edge of π'_1 be $x_k \geqslant b_0 + \sum_{j' \in [n]} b_{j'} x_{j'}$. Then similarly to the case that G_φ contains an edge of weight greater than 1, we utilize π'_1 to apply the replacements to the inequality $x_i \geqslant b_0 + \sum_{j' \in [n]} b_{j'} x_{j'}$. Let the resulting inequality be $Ineq' \equiv x_k \geqslant c_0 + \sum_{j' \in [n]} c_{j'} x_{j'}$. From the fact that π'_1 is a path from x_k to x_k, we know that $c_k \geqslant 1$. Moreover, $c_{j_2} \geqslant 1$ since x_i is replaced by $a'_0 + \sum_{j' \in [n]} a'_{j'} x_{j'}$ at least once and $a'_{j_2} = a_2 \geqslant 1$. Then we utilize π_2 to apply the replacements to the inequality $Ineq'$, and obtain the inequality $Ineq'' \equiv x_k \geqslant d_0 + \sum_{j' \in [n]} d_{j'} x_{j'}$. Then from $c_k \geqslant 1$ and $c_{j_2} \geqslant 1$, we deduce that $d_k \geqslant 2$. Therefore, $-d_0 \geqslant (d_k - 1)x_k + \sum_{j' \in [n], j \neq k} d_{j'} x_{j'}$. This means that if φ is satisfiable, then the value of x_k is bounded by $-c_0$.

Consequently, in this case, there is a constant C such that the values of variables in every solution are bounded by C. □

Therefore, from Proposition 2, if G_φ contains an edge of weight greater than 1 or two distinct edges out of some vertex x_i of the same label, then the number of solutions satisfying φ is finite and we can find the minimum solution by guessing the solution candidates from the least to the greatest one by one and check whether they are indeed solutions.

In the sequel, let us assume that *the weight of each edge in G_φ is 1 and no two edges out of the same vertex are of the same label.* Then for each vertex x_i, we guess all the simple cycles that involve x_i and do the following for each such simple cycle. Let $C = x_{l_1} \cdots x_{l_r} x_{l_{r+1}}$ be a simple cycle where $l_{r+1} = l_1 = i$. We compute $u_{C,l_1}, \cdots, u_{C,l_r}$ as follows. For each $j' \in [r]$, let the inequality corresponding to the j'-th edge of C be $x_{l_{j'}} \geqslant b_{l_{j'}} + x_{l_{(j' \bmod r)+1}}$.

- If $\sum_{j' \in [r]} b_{l_{j'}} > 0$, then φ' is unsatisfiable.
- Otherwise, for each $j' \in [r]$, let $u_{C,l_{j'}} := \max(0, b_{l_{j'}}, b_{l_{j'}} + b_{l_{j'+1}}, \cdots, b_{l_{j'}} + \cdots + b_{l_{(j'+r-3) \bmod r+1}})$.

Finally, for each $i \in [n]$, we compute the minimum solution as $(u_1, \cdots, u_n)$ such that for each $i \in [n]$, u_i is the maximum of all u_{C,l_1} such that $C = x_{l_1} \cdots x_{l_r} x_{l_{r+1}}$ is a simple cycle where $l_{r+1} = l_1 = i$.

More general case. Suppose that G_φ is neither acyclic nor strongly connected. Let $\mathcal{S}_{G_\varphi}$ be the SCC graph of G_φ. We compute a topological sorting of $\mathcal{S}_{G_\varphi}$, say $C_1, \cdots, C_k$. Let $j := k$. We iterate the following procedure, until $j = 1$.

1. Suppose $u_{j'} \in \mathbb{N}$ has been computed for each $x_{j'} \in V(C_{j+1}) \cup \cdots \cup V(C_k)$. For each $x_{j''} \in V(C_j)$, replace each occurrence of $x_{j'} \in V(C_{j+1}) \cup \cdots \cup V(C_k)$ in the right-hand sides of the inequalities of $x_{j''}$ with $u_{j'}$. Let ψ_{C_j} be the conjunction of the resulting inequalities of $x_{j''} \in V(C_j)$. Then we can utilize the aforementioned procedure for strongly connected dependency graphs to compute the minimum solution $(u_{j''})_{x_{j''} \in V(C_j)}$ for ψ_{C_j}. If during the computation, ψ_{C_j} is discovered to be unsatisfiable, then φ is unsatisfiable and the computation stops.
2. Let $j := j - 1$.

4 Discussions

In the width inference problem, we have not considered the expressions of the form $shl(x, y)$, where y is an unsigned integer. It is left as the future work.